Find the Error

Grade 4 · 4.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Ann has 3 stickers. Ben has 4 times as many. How many does Ben have? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 4 instead of multiplying. Times as many means 4 × 3 = 12.**

Correct answer: **12 stickers**

Find the Error

Grade 4 · 4.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rope is 5 meters long. Another rope is 2 times as long. How long is it? Student's answer: 7 meters

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2 to 5. Two times as long means 2 × 5 = 10.**

Correct answer: **10 meters**

Find the Error

Grade 4 · 4.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Tom has 12 cards. Sam has 3 cards. How many times as many does Tom have? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted. To find how many times as many, divide 12 by 3 = 4.**

Correct answer: **4 times as many**

Find the Error

Grade 4 · 4.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Ana has 8 books. Ben has 3 more than Ana. How many does Ben have? Student's answer: 24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied instead of adding. 3 more than 8 is 8 + 3 = 11.**

Correct answer: **11 books**

Find the Error

Grade 4 · 4.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Ana has 8 books. Ben has 3 times as many. How many does Ben have? Student's answer: 11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added instead of multiplying. 3 times as many is 3 × 8 = 24.**

Correct answer: **24 books**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Tia has 15 shells. Max has 5 fewer. How many does Max have? Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added. Fewer means subtracting: 15 - 5 = 10.**

Correct answer: **10 shells**

Find the Error

Grade 4 · 4.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A store has 4 boxes with 12 pencils each. 9 are sold. How many are left? Student's answer: 57

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 9 to the total. 4 × 12 = 48, then 48 − 9 = 39.**

Correct answer: **39 pencils**

Find the Error

Grade 4 · 4.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Mia has 3 bags with 7 marbles each and 4 more marbles. How many in all? Student's answer: 84

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the extra marbles. 3 × 7 = 21, then 21 + 4 = 25.**

Correct answer: **25 marbles**

Find the Error

Grade 4 · 4.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

45 apples are shared equally among 5 baskets. Each basket gets 2 more apples. How many in each basket? Student's answer: 47

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the 45 apples instead of dividing. 45 ÷ 5 = 9, then 9 + 2 = 11.**

Correct answer: **11 apples**

Find the Error

Grade 4 · 4.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 15 a multiple of 3? Student's answer: no

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student checked the wrong way. 3 × 5 = 15, so 15 is a multiple of 3.**

Correct answer: **Yes**

Find the Error

Grade 4 · 4.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

List the factors of 12. Student's answer: 1, 2, 3, 4, 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student left out 12. Every number has itself as a factor, so 12 is also a factor.**

Correct answer: **1, 2, 3, 4, 6, 12**

Find the Error

Grade 4 · 4.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 7 prime? Student's answer: no

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a mistake. 7 has only two factors, 1 and 7, so it is prime.**

Correct answer: **Yes, 7 is prime.**

Find the Error

Grade 4 · 4.OA.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Pattern starts at 7 and adds 5: 7, 12, 17, 23. Find the error.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student wrote 23 instead of 22. The rule adds 5 each time, so 17 + 5 = 22.**

Correct answer: **22 should replace 23**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Pattern starts at 2 and multiplies by 3: 2, 6, 18, 36. Find the error.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 18 by 2 instead of 3. 18 × 3 = 54.**

Correct answer: **54 should replace 36**

Find the Error

Grade 4 · 4.OA.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says the rule for 3, 6, 9, 12 is 'add 4.'

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 4 instead of 3. The rule is add 3.**

Correct answer: **Add 3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In 500, the digit 5 is worth how much? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave the digit, not its place value. The 5 is in the hundreds place, so it is worth 500.**

Correct answer: 500

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

10 times 300 = ? Student's answer: 30

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not shift the digits. Multiplying by 10 moves each digit one place to the left.**

Correct answer: **3,000**

Find the Error

Grade 4 · 4.NBT.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many tens are in 4 hundreds? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted the hundreds. Each hundred has 10 tens, so 4 hundreds is 40 tens.**

Correct answer: **40 tens**

Find the Error

Grade 4 · 4.NBT.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 'four thousand twenty' as a numeral... Student's answer: 4,020 and says 'four thousand seventy-two' is 4,020.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student mixed up the two numbers. Four thousand seventy-two is 4,072.**

Correct answer: 4,072

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 'eighty-two thousand fifty-one' as a numeral. Student's answer: 82,501

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student wrote the digits in the wrong places. Fifty-one is 051 at the end, so the number is 82,051.**

Correct answer: **82,051**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare 4,523 and 4,532. Student writes: 4,523 > 4,532

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student reversed the symbol. Both have 4 thousands and 5 hundreds, but 2 tens is less than 3 tens.**

Correct answer: **4,523 < 4,532**

Find the Error

Grade 4 · 4.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 3,865 to the nearest hundred. Student's answer: 3,800

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 3,865 is closer to 3,900.**

Correct answer: **3,900**

Find the Error

Grade 4 · 4.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 47,650 to the nearest thousand. Student's answer: 47,000

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 47,650 is halfway or more, so it rounds up to 48,000.**

Correct answer: **48,000**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 638 to the nearest ten. Student's answer: 630

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 638 is closer to 640.**

Correct answer: **640**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

4,326 + 2,518 = ? Student's answer: 6,734

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made an addition error in the tens or hundreds. 4,326 + 2,518 = 6,844.**

Correct answer: 6,844

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

7,000 - 3,456 = ? Student's answer: 4,456

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a regrouping error. 7,000 - 3,456 = 3,544.**

Correct answer: 3,544

Find the Error

Grade 4 · 4.NBT.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

5,678 - 2,345 = ? Student's answer: 3,343

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a subtraction error in the ones place. 5,678 - 2,345 = 3,333.**

Correct answer: 3,333

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

6 × 2,000 = ? Student's answer: 8,000

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added instead of multiplying. 6 × 2 = 12, so 6 × 2,000 = 12,000.**

Correct answer: **12,000**

Find the Error

Grade 4 · 4.NBT.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

23 × 4 = ? Student's answer: 82

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a regrouping error. 4 × 3 = 12 and 4 × 20 = 80, so 23 × 4 = 92.**

Correct answer: 92

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

17 × 25 = ? Student's answer: 325

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student left out a partial product. 17 × 25 = 425.**

Correct answer: **425**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

56 ÷ 4 = ? Student's answer: 13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a division error. 4 × 14 = 56, so 56 ÷ 4 = 14.**

Correct answer: 14

Find the Error

Grade 4 · 4.NBT.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

75 ÷ 6 = ? Student's answer: 12 R 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a remainder error. 6 × 12 = 72, and 75 − 72 = 3.**

Correct answer: **12 R 3**

Find the Error

Grade 4 · 4.NBT.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

96 ÷ 8 = ? Student's answer: 11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a division error. 8 × 12 = 96, so 96 ÷ 8 = 12.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $\frac{2}{3}$ equal to $\frac{4}{6}$? Student's answer: no

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not multiply both parts by 2. $\frac{2}{3} = \frac{4}{6}$.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find a fraction equal to $\frac{3}{4}$ with denominator 8. Student's answer: $\frac{3}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not change the numerator. Multiply both parts by 2 to get $\frac{6}{8}$.**

Correct answer: $\frac{6}{8}$

Find the Error

Grade 4 · 4.NF.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $\frac{3}{5}$ equal to $\frac{6}{15}$? Student's answer: yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not check the multiplication. $3 \times 3 = 9$, so $\frac{3}{5} = \frac{9}{15}$, not $\frac{6}{15}$.**

Correct answer: **No. $\frac{3}{5} = \frac{9}{15}$.**

Find the Error

Grade 4 · 4.NF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\frac{3}{8}$ and $\frac{5}{8}$. Student writes: $\frac{3}{8} > \frac{5}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student reversed the symbol. With the same denominator, $\frac{3}{8}$ is less than $\frac{5}{8}$.

Correct answer: $\frac{3}{8} < \frac{5}{8}$

Find the Error

Grade 4 · 4.NF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\frac{2}{3}$ and $\frac{3}{4}$. Student writes: $\frac{2}{3} > \frac{3}{4}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student reversed the comparison. $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $\frac{2}{3}$ is less.

Correct answer: $\frac{2}{3} < \frac{3}{4}$

Find the Error

Grade 4 · 4.NF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\frac{1}{2}$ and $\frac{5}{10}$. Student writes: $\frac{1}{2} < \frac{5}{10}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the wrong symbol. $\frac{1}{2}$ and $\frac{5}{10}$ are equal fractions.**

Correct answer: $\frac{1}{2} = \frac{5}{10}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write $\frac{3}{5}$ as a sum of unit fractions. Student's answer: $\frac{1}{5} + \frac{2}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used a non-unit fraction. A unit fraction has a numerator of 1, so $\frac{3}{5} = \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$.**

Correct answer: $\frac{1}{5} + \frac{1}{5} + \frac{1}{5}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\frac{3}{4} + \frac{1}{4} = ?$ Student's answer: $\frac{4}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the denominators. When denominators match, keep the denominator: $\frac{4}{4} = 1$.**

Correct answer: 1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\frac{5}{6} - \frac{2}{6} = ?$ Student's answer: $\frac{3}{0}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted the denominators. Keep the denominator: $\frac{5}{6} - \frac{2}{6} = \frac{3}{6}$, which simplifies to $\frac{1}{2}$.**

Correct answer: $\frac{3}{6} = \frac{1}{2}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$3 \times \frac{1}{4} = ?$ Student's answer: $\frac{3}{12}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the denominator too. Multiply only the numerator: 3 × 1 = 3, so the answer is $\frac{3}{4}$.**

Correct answer: $\frac{3}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$4 \times \frac{2}{5} = ?$ Student's answer: $\frac{6}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2 instead of multiplying. $4 \times 2 = 8$, so the answer is $\frac{8}{5}$.**

Correct answer: $\frac{8}{5}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$5 \times \frac{1}{8} = ?$ Student's answer: $\frac{5}{40}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the denominator. Keep the denominator: $\frac{5}{8}$.**

Correct answer: **$\frac{5}{8}$**

Find the Error

Grade 4 · 4.NF.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write $\frac{3}{10}$ as hundredths. Student's answer: $\frac{3}{100}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student changed the denominator without the numerator. Multiply both by 10: $\frac{30}{100}$.**

Correct answer: $\frac{30}{100}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write $\frac{7}{10}$ as hundredths. Student's answer: $\frac{70}{10}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student kept the denominator at 10. To get hundredths, multiply both by 10: $\frac{70}{100}$.**

Correct answer: **$\frac{70}{100}$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.4 as a fraction. Student's answer: $\frac{4}{100}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the wrong place value. The 4 is in the tenths place, so 0.4 = $\frac{4}{10}$.**

Correct answer: $\frac{4}{10}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.7 as a fraction. Student's answer: $\frac{7}{100}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used hundredths. The 7 is in the tenths place, so 0.7 = $\frac{7}{10}$.**

Correct answer: $\frac{7}{10}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write $\frac{4}{10}$ as a decimal. Student's answer: 0.04

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the wrong place value. 4 tenths is 0.4.**

Correct answer: **0.4**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.52 in expanded form. Student's answer: 5 + 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not show the place values. 0.52 = $\frac{5}{10} + \frac{2}{100}$.**

Correct answer: $\frac{5}{10} + \frac{2}{100}$

Find the Error

Grade 4 · 4.NF.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare 0.4 and 0.37. Student writes: 0.4 < 0.37

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student compared the digits from the wrong side. 0.4 is 0.40, which is greater than 0.37.**

Correct answer: **0.4 > 0.37**

Find the Error

Grade 4 · 4.NF.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare 0.6 and 0.60. Student writes: 0.6 > 0.60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student thought the extra zero made a difference. 0.6 and 0.60 are equal.**

Correct answer: **0.6 = 0.60**

Find the Error

Grade 4 · 4.NF.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare 0.25 and 0.5. Student writes: 0.25 > 0.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student compared 25 and 5 as whole numbers. 0.5 is 0.50, which is greater than 0.25.**

Correct answer: **0.25 < 0.5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

3 feet = ? inches. Student's answer: 3 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student kept the same number. There are 12 inches in 1 foot, so 3 feet = 36 inches.**

Correct answer: **36 inches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

2 meters = ? centimeters. Student's answer: 20 cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 10 instead of 100. There are 100 cm in 1 meter, so 2 meters = 200 cm.**

Correct answer: **200 cm**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

5 kg = ? grams. Student's answer: 500 g

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 100 instead of 1,000. There are 1,000 g in 1 kg, so 5 kg = 5,000 g.**

Correct answer: **5,000 g**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A ribbon is 6 feet long. Ben uses 18 inches. How many inches are left? Student's answer: 18 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave the amount used. 6 feet is 72 inches, and 72 - 18 = 54 inches left.**

Correct answer: **54 inches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bottle holds 2 L 300 mL. It gives away 500 mL. How much is left? Student's answer: 2 L 200 mL

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted the liters only. 2,300 mL − 500 mL = 1,800 mL, or 1 L 800 mL.**

Correct answer: **1 L 800 mL**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$3.45 + $2.80 = ? Student's answer: $5.25

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made an addition error. $3.45 + $2.80 = $6.25.**

Correct answer: **$6.25**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a rectangle 9 by 4. Student's answer: 26 square units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the perimeter. Area is 9 × 4 = 36 square units.**

Correct answer: **36 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the perimeter of a rectangle 7 by 3. Student's answer: 21

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the sides. Perimeter is 7 + 3 + 7 + 3 = 20.**

Correct answer: **20 units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the perimeter of a square with side 5. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only two sides. Perimeter is 5 + 5 + 5 + 5 = 20.**

Correct answer: **20 units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot has 3 X's at $\frac{1}{2}$ inch and 2 X's at $\frac{3}{4}$ inch. Total length? Student's answer: $2\ \frac{1}{4}$ inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the values wrong. $3 \times \frac{1}{2} = 1\ \frac{1}{2}$, and $2 \times \frac{3}{4} = 1\ \frac{1}{2}$, so the total is 3 inches.**

Correct answer: **3 inches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

4 X's are each $\frac{1}{8}$ inch. Total length? Student's answer: $\frac{4}{64}$ inch

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the denominators. Keep the denominator: $4 \times \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$ inch.**

Correct answer: **$\frac{1}{2}$ inch**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot has 2 X's at $\frac{1}{4}$ inch and 1 X at $\frac{3}{8}$ inch. Total length? Student's answer: $\frac{3}{12}$ inch

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student multiplied the denominators. $\frac{2}{4} + \frac{3}{8} = \frac{4}{8} + \frac{3}{8} = \frac{7}{8}$ inch.

Correct answer: $\frac{7}{8}$ inch

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 100° angle is acute or obtuse? Student's answer: acute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student confused the terms. Acute angles are less than 90°, and 100° is more than 90°, so it is obtuse.**

Correct answer: **Obtuse**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right angle measures how many degrees? Student's answer: 180°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave a straight angle. A right angle measures 90°.**

Correct answer: 90°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 45° angle is a right angle. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student confused a right angle with an acute angle. 45° is less than 90°, so it is acute.**

Correct answer: **No. 45° is an acute angle.**

Find the Error

Grade 4 · 4.MD.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An angle's ray lines up with the 0 on the inner scale and the other ray points to 50 on the inner scale. Student reads 130°.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student read the outer scale. Use the scale that starts at 0 on the same ray, which gives 50°.**

Correct answer: **50°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student draws a 90° angle but says it measures 45°.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student misread the protractor. A 90° angle is a right angle, and it measures 90°.**

Correct answer: **No. The angle measures 90°.**

Find the Error

Grade 4 · 4.MD.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student measures an angle and reads 75°, then says it is 105° because they read the other scale.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student read the wrong scale. Use the scale that starts at 0 on the ray that lines up with the protractor's base.**

Correct answer: **75°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Angle parts are 30° and 45°. What is the total? Student's answer: 65°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made an addition error. 30° + 45° = 75°.**

Correct answer: 75°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles together make 90°, and one is 35°. What is the other? Student's answer: 65°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added instead of subtracting. 90° − 35° = 55°.**

Correct answer: **55°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 120° angle is split into 50° and 60°. Student says this is correct.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The parts add to 110°, not 120°. A 10° part is missing.**

Correct answer: **No. The parts add to 110°, so 10° is missing.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a ray has two endpoints. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A ray starts at one endpoint and goes on forever in one direction.**

Correct answer: **No. A ray has one endpoint.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says parallel lines meet at one point.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Parallel lines never meet. They stay the same distance apart.**

Correct answer: **No. Parallel lines never meet.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says perpendicular lines form a 45° angle.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Perpendicular lines form right angles, which measure 90°.**

Correct answer: **No. Perpendicular lines form 90° angles.**

Find the Error

Grade 4 · 4.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a triangle with one right angle is an equilateral triangle.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **An equilateral triangle has three equal sides and no right angle. A triangle with one right angle is a right triangle.**

Correct answer: **A right triangle**

Find the Error

Grade 4 · 4.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says an acute triangle has one obtuse angle.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **An acute triangle has all three angles acute. A triangle with one obtuse angle is an obtuse triangle.**

Correct answer: **No. All angles in an acute triangle are acute.**

Find the Error

Grade 4 · 4.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a rectangle has 4 equal sides.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A rectangle has 2 pairs of equal sides. Only a square has 4 equal sides.**

Correct answer: **No. A rectangle has 2 pairs of equal sides.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many lines of symmetry does a square have? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted only the vertical and horizontal lines. A square also has 2 diagonal lines, for 4 in all.**

Correct answer: 4

Find the Error

Grade 4 · 4.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many lines of symmetry does a rectangle that is not a square have? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted as if it were a square. A non-square rectangle has 2 lines of symmetry.**

Correct answer: 2

Find the Error

Grade 4 · 4.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many lines of symmetry does an equilateral triangle have? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student missed two lines. An equilateral triangle has 3 lines of symmetry.**

Correct answer: 3